Multivariate Calculus Lecture Notes
Section 15.6 – Change of Variables

I. Substitutions in double integrals
   a. In multivariable calculus when we want to perform a change of variables we generally start with an integral like $\iint_R f(x,y)\,dx\,dy$ and do a substitution with $x = x(u,v)$, $y = y(u,v)$.
   b. Example: Rectangular to polar coordinates
   c. In a single integral, $dx = g'(u)du$, so the change of variables entails an extra factor of $g'(u)$.
   d. In a multiple integral, we have $dx\,dy = |J(u,v)|du\,dv$, provided the mapping $x = x(u,v)$, $y = y(u,v)$ is one-to-one.
      i. $J(u,v)$ is called the *Jacobian* of the transformation from (x,y) to (u,v).
      ii. $J(u,v)$ is given by $J(u,v) = \dfrac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix}$.
      iii. The derivation of this formula is beyond the scope of this course.
   e. Thus $\iint_R f(x,y)\,dx\,dy = \iint_S f(x(u,v),y(u,v))|J(u,v)|du\,dv$, where S is the transformed region of integration.

II. Examples
   a. Rectangular to polar
      i. The Jacobian is $J(r,\theta) = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r$.
      ii. Thus $dx\,dy = r\,dr\,d\theta$.
   b. Evaluate the integral $\iint_R (2y+x)dx\,dy$ for the region in the first quadrant bounded by $y = -2x+4$, $y = -2x+7$, $y = x-2$, and $y = x+1$.
      i. Region of integration:

      ii. The region is a diamond shape that is neither vertically nor horizontally simple, so let's use the bounds to obtain a transformation.
         1. Rewriting two of the bounds as $y+2x = 4$ and $y+2x = 7$ means that letting $u = y+2x$ will give us constant bounds of 4 and 7 for u.
         2. Similarly, letting $v = y-x$ will give us constant bounds of -2 and 1 for v.
      iii. Under this transformation, the new region of integration is as follows:

      iv. We need to solve for x and y in terms of u and v to finish the problem.

1. Subtracting the v equation from the u equation yields $u - v = 3x$.
2. Thus $x = \frac{1}{3}(u - v)$.
3. And $v = y - \frac{1}{3}(u - v) \implies y = \frac{1}{3}u + \frac{2}{3}v$.

v. The integrand then becomes $2y + x = u + v$.

vi. And the Jacobian is $J(u, v) = \begin{vmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{vmatrix} = \frac{2}{9} + \frac{1}{9} = \frac{1}{3}$.

vii. Thus the transformed integral is

$$\frac{1}{3}\int_4^7 \int_{-2}^1 (u+v)\,dv\,du = \frac{1}{3}\int_4^7 \left(uv + \frac{1}{2}v^2\Big|_{-2}^1\right) du = \frac{1}{3}\int_4^7 \left(3u - \frac{3}{2}\right) du$$
$$= \frac{1}{3}\left[\frac{3}{2}u^2 - \frac{3}{2}u\right]_4^7 = \frac{1}{3}\left(\frac{147}{2} - \frac{21}{2} - \frac{48}{2} + \frac{12}{2}\right) = \frac{45}{3} = 15.$$

c. Evaluate the integral $\iint_R e^{(x+y)/(x-y)}\,dA$, where R is the trapezoidal region with vertices $(1, 0)$, $(2, 0)$, $(0, -2)$, and $(0, -1)$.

i. Region of integration

1. Graphically:

2. This is not vertically or horizontally simple.
3. The bounds on R are $y = 0$, $x - y = 2$, $x = 0$, and $x - y = 1$.

ii. Since the integrand is so ugly, an obvious substitution to try is $u = x + y$, $v = x - y$.

iii. Finding the transformed region of integration

1. Boundaries
   - The boundary $y = 0$ becomes $u = v$.
   - The boundary $x - y = 2$ becomes $v = 2$.
   - The boundary $x = 0$ becomes $u = -v$.
   - The boundary $x - y = 1$ becomes $v = 1$.
2. Graphically:

iv. Finding the Jacobian

1. Solving for x and y yields
   - $x = \frac{1}{2}u + \frac{1}{2}v$
   - $y = \frac{1}{2}u - \frac{1}{2}v$.
2. The Jacobian is thus $J(u, v) = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$.

v. Therefore,

$$\iint_R e^{(x+y)/(x-y)}\,dA = \int_1^2 \int_{-v}^{v} \frac{1}{2}e^{u/v}\,du\,dv = \frac{1}{2}\int_1^2 ve^{u/v}\Big|_{-v}^{v}\,dv = \frac{1}{2}\int_1^2 v(e-e^{-1})\,dv$$
$$= \frac{1}{4}(e-e^{-1})v^2\Big|_1^2 = \frac{1}{4}(e-e^{-1})(4-1) = \frac{3}{4}(e-e^{-1}).$$

III. Interpretation of the Jacobian

a. Suppose we are integrating $\iint_R dx\,dy$.

b. This gives the area of region R.

c. When we make a variable change we have $\iint_R dx\,dy = \iint_S |J(u,v)|\,du\,dv$.

d. Now, $\iint_S du\,dv$ by itself would be the area of S.

e. Thus $|J(u,v)|$ is the factor by which the area around the point (u,v) in S contracts or expands when we transform it to R.

f. And $J(u,v)$ is negative if the transformation from S to R reverses the orientation of S.

g. Examples

i. If the transformation is linear, so that $|J(u,v)|$ is constant, the area of R is 4, and the area of S is 12, then $|J(u,v)| = \frac{1}{3}$.

ii. If we are mapping the unit square to the unit square via the transformation $u = 1-x$, $v = y$, then $J(u,v) = \begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} = -1$, and we see the orientation reversal:

IV. Substitutions in triple integrals

a. When you substitute using triple integrals, you switch from (x,y,z) to some three-variable coordinate system (u,v,w), using functions like $x = g(u,v,w), y = h(u,v,w), z = k(u,v,w)$ to do the transformation.

b. The formula is

$$\iiint_D f(x,y,z)\,dx\,dy\,dz = \iiint_Q f(g(u,v,w),h(u,v,w),k(u,v,w))\big|J(u,v,w)\big|\,du\,dv\,dw.$$

c. For functions of three variables, the Jacobian is

$$J(u,v,w) = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} & \dfrac{\partial x}{\partial w} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} & \dfrac{\partial y}{\partial w} \\ \dfrac{\partial z}{\partial u} & \dfrac{\partial z}{\partial v} & \dfrac{\partial z}{\partial w} \end{vmatrix}.$$

i. Alternatively,

$$J(u,v,w) = \frac{\partial x}{\partial u}\left(\frac{\partial y}{\partial v}\frac{\partial z}{\partial w} - \frac{\partial z}{\partial v}\frac{\partial y}{\partial w}\right) - \frac{\partial x}{\partial v}\left(\frac{\partial y}{\partial u}\frac{\partial z}{\partial w} - \frac{\partial z}{\partial u}\frac{\partial y}{\partial w}\right) + \frac{\partial x}{\partial w}\left(\frac{\partial y}{\partial u}\frac{\partial z}{\partial v} - \frac{\partial z}{\partial u}\frac{\partial y}{\partial v}\right).$$

ii. This is similar to the cross product formula.

d. Examples

i. Rectangular to cylindrical (similar to rectangular to polar switch)

ii. Rectangular to spherical (homework problem)

iii. Find the volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

1. This will be ugly in rectangular, cylindrical, or spherical coordinates.
2. However, the change of variables $u = \frac{x}{a}$, $v = \frac{y}{b}$, $w = \frac{z}{c}$, will transform the ellipsoid into a sphere of radius 1.
3. We have $x = au$, $y = bv$, $z = cw$.
4. The Jacobian is $J(u,v,w) = \begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = abc.$
5. Thus, if S is the original ellipsoid in xyz-space and Q is the unit sphere in uvw-space, the volume of the ellipsoid is given by

$$\iiint_S dx\,dy\,dz = \iiint_Q abc\,du\,dv\,dw = abc\iiint_Q du\,dv\,dw = \frac{4}{3}\pi abc.$$

Homework: Section 15.6 #1, 13, 16, 21, 31, 32, 35, 38, 43.